

Planets Database

AIM

To write and execute a **Prolog program to create a database containing information about planets and retrieve the required information.**

ALGORITHM

1. Start the program.
2. Define facts containing the names of planets and their properties.
3. Store information such as the planet's position, type, and number of moons.
4. Use Prolog queries to retrieve information from the database.
5. Display the required planet information.
6. Stop the program.

PROGRAM

% Prolog program for Planets Database

planet(mercury, terrestrial, 0).

planet(venus, terrestrial, 0).

planet(earth, terrestrial, 1).

planet(mars, terrestrial, 2).

planet(jupiter, gas_giant, 95).

planet(saturn, gas_giant, 146).

planet(uranus, ice_giant, 28).

planet(neptune, ice_giant, 16).

OUTPUT

```
14 ?- consult(['C:/Users/Lokesh/Desktop/17-33/planets.pl']).
true.

15 ?- planet(earth,Type,Moons).
Type = terrestrial,
Moons = 1.

16 ?- planet(Planet,Type,0).
Planet = mercury,
Type = terrestrial .

17 ?- planet(Planet,gas_giant,Moons).
Planet = jupiter,
Moons = 95 ▲
```

RESULT

Thus, the **Prolog** program to create and retrieve information from a **Planets Database** was **successfully executed**, and the required planet information was obtained successfully.